

Mathematical and Logical Foundations of Computer Science

Propositional Logic - Additional Examples

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Propositions

Upper case letters stand for variables, while lower case letters stand for atomic propositions. We will often use variables rather than specific atomic propositions.

Here are examples of propositions:

- ▶ $P \rightarrow Q \rightarrow R$
 - ▶ it stands for $P \rightarrow (Q \rightarrow R)$
 - ▶ because $\rightarrow$ is right-associative
 - ▶ this is different from $(P \rightarrow Q) \rightarrow R$
 - ▶ it uses 2 variables P and Q
- ▶ $p \rightarrow q \rightarrow r$
 - ▶ it stands for $p \rightarrow (q \rightarrow r)$
 - ▶ because $\rightarrow$ is right-associative
 - ▶ this is different from $(p \rightarrow q) \rightarrow r$
 - ▶ it uses 2 atoms p and q

Propositions

Here are further examples of propositions:

- ▶ $P \wedge Q \rightarrow P \vee Q$
 - ▶ it stands for $(P \wedge Q) \rightarrow (P \vee Q)$
 - ▶ because $\wedge$ and $\vee$ have precedence over $\rightarrow$
 - ▶ this is different from $P \wedge (Q \rightarrow P \vee Q)$
 - ▶ it uses 2 variables P and Q
- ▶ $\neg\neg P \rightarrow P$
 - ▶ it stands for $(\neg(\neg P)) \rightarrow P$
 - ▶ because $\neg$ has precedence over $\rightarrow$
 - ▶ this is different from $\neg\neg(P \rightarrow P)$
 - ▶ it uses 1 variable P

Propositions

Those are not propositions:

- ▶ $P \wedge \wedge Q$

- ▶ this is not derivable from the syntax of propositional logic
- ▶ it is not possible to have two $\wedge$ next to each other

- ▶ $P \rightarrow \wedge Q$

- ▶ this is not derivable from the syntax of propositional logic
- ▶ it is not possible to have a $\wedge$ next to a $\rightarrow$

Natural Deduction Proofs

Here is a Natural Deduction proof of $A \vee B \vdash C \rightarrow (A \vee B) \wedge C$

$$\frac{\frac{A \vee B \quad \overline{C}^1}{(A \vee B) \wedge C} [\wedge I]}{C \rightarrow (A \vee B) \wedge C}^1 [\rightarrow I]$$

Note that $C \rightarrow (A \vee B) \wedge C$ is read as $C \rightarrow ((A \vee B) \wedge C)$

Natural Deduction Proofs

Here is a Natural Deduction proof of $Q \vdash (Q \rightarrow R) \rightarrow R$

$$\frac{\frac{\overline{Q \rightarrow R}^1 \quad Q}{R} [\rightarrow E]}{(Q \rightarrow R) \rightarrow R}^1 [\rightarrow I]$$

Natural Deduction Proofs

Here is a Natural Deduction proof of $\vdash \neg(A \wedge B) \rightarrow A \rightarrow \neg B$

$$\frac{\frac{\frac{}{\neg(A \wedge B)} 1 \quad \frac{\frac{A}{\neg A} 2 \quad \frac{B}{\neg B} 3}{A \wedge B} [\wedge I]}{\perp} [\neg E] \quad \frac{}{\neg B} 3 [\neg I]}{A \rightarrow \neg B} 2 [\rightarrow I] \quad \frac{}{\neg(A \wedge B) \rightarrow A \rightarrow \neg B} 1 [\rightarrow I]$$

Note that $\neg(A \wedge B) \rightarrow A \rightarrow \neg B$ is read as $\neg(A \wedge B) \rightarrow (A \rightarrow \neg B)$